

Silverline House — 25-Storey Residential HRB

Rainscreen Cladding Specification

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1 Introduction

The rainscreen cladding system has been selected and specified to comply with the requirements of Approved Document B (external fire spread), Approved Document C (resistance to weather), and Approved Document L (energy efficiency). The system comprises fibre cement panels in an open-joint configuration, a ventilated cavity, mineral wool insulation boards, an air and vapour control layer (AVCL), and the primary wall construction. The system has been tested in accordance with BS 8414-1 and assessed against BR 135 Fire Performance of External Thermal Insulation for Walls of Multistorey Buildings. Third-party certification is provided by the BBA. This document forms part of the Gateway 2 application for Building Control Approval submitted to the Building Safety Regulator under the Building Safety Act 2022. The information presented herein has been prepared in accordance with the CLC Guidance Suite for new Higher-Risk Buildings and the requirements set out in Guidance Note 04 (Application Information Schedule) and Guidance Note 06 (Document Management and Submission). All design work described in this document has been carried out by suitably qualified engineers and specialists with relevant experience in higher-risk residential buildings.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 System Components and Performance

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3 Installation Requirements

Below-ground waterproofing is provided by a Type C drained protection system in accordance with BS 8102:2022, comprising a primary tanked membrane applied to the external face of the basement retaining walls and underside of the ground floor slab, supplemented by a drainage layer and sump pump system. The membrane is a hot-applied reinforced bituminous sheet system with a minimum thickness of 4 mm applied to prepared concrete substrate. Lap seals are formed using a heat torch and are inspected and pressure-tested prior to backfilling. The drainage layer consists of a geocomposite drainage sheet with a geotextile filter fabric bonded to the soil-facing surface, discharging to perimeter drainage sumps. Penetrations through the tanking system for services entries are sealed using proprietary puddle flanges and sleeve systems with BBA certification. The primary rainscreen cladding system is a rear-ventilated facade with horizontal support rails at 600 mm vertical centres attached to stainless steel L-brackets fixed at 900 mm horizontal centres to the structural slab edge. The bracket design must accommodate a slab-edge to finished face dimension of 250 mm, a maximum vertical load per bracket of 3.2 kN, and a maximum horizontal (wind) load per bracket of 1.8 kN. Thermal break pads are provided between each bracket foot and the concrete slab edge to limit thermal bridging. Adjustment slots in the bracket allow for construction tolerances of ± 15 mm in all three axes. The rail-to-bracket connection is via stainless steel M12 bolts with load-spreading plates, designed to resist the combined effects of dead load, wind load, and seismic actions.

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